



E-Commerce Sales Data Analysis and Reporting System

Business Insights Report

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1 Introduction

1.1 Background

The e-commerce company operates across multiple product categories including Electronics, Clothing, Home & Kitchen, Furniture, and Sports. Management requires a consolidated view of customer behaviour, product performance, and sales trends to make data-driven decisions.

However, the data was stored across four separate CSV files, containing missing values, duplicates, and inconsistencies. This made it difficult to generate meaningful reports and insights.

1.2 Objectives

The primary objectives of this project are:

1. Load and integrate data from multiple CSV files.
2. Validate and clean the data to ensure quality.
3. Transform and merge the data into a single analytical dataset.
4. Compute key business metrics and KPIs.
5. Perform customer segmentation using RFM analysis.
6. Estimate customer lifetime value (CLV) using statistical archetypes.
7. Generate 10 actionable business insights.
8. Export the final dataset and analysis outputs.

1.3 Scope and Limitations

Included:

- All 4 provided datasets were analysed.
- Data cleaning, validation, transformation, and integration.
- RFM customer segmentation with quintile scoring.
- Statistical CLV projection using rule-based archetypes.
- Cohort retention analysis.
- Pivot tables and multi-level aggregations.

Excluded:

- Predictive machine learning models.
- Real-time data processing.
- External data sources (marketing spend, competitor data, etc.).
- Profit margin or cost analysis (not available in the data).

1.4 Assumptions

- All orders are equally important.
- The 12-month period covered is representative of annual behaviour.
- Future behaviour will resemble historical patterns.
- Customers with no orders are excluded from RFM analysis.
- CLV is projected as total spend only (no margin adjustment).
- Customer tenure is calculated dynamically using today's date, making the pipeline reproducible and independent of dataset reference dates.

2 Methodology

2.1 Data Sources

The analysis was based on four CSV files provided by the e-commerce company. Each file represents a different aspect of the business operations.

Table 1: Dataset Descriptions

Dataset	Key Columns	Rows
Customers	customer_id, customer_name, city, signup_date	201
Products	product_id, product_name, category, price	50
Orders	order_id, customer_id, order_date, region	1000
Order_Items	order_id, product_id, quantity	2501

2.2 ETL Pipeline Architecture

The complete data processing pipeline follows a standard Extract-Transform-Load (ETL) architecture.

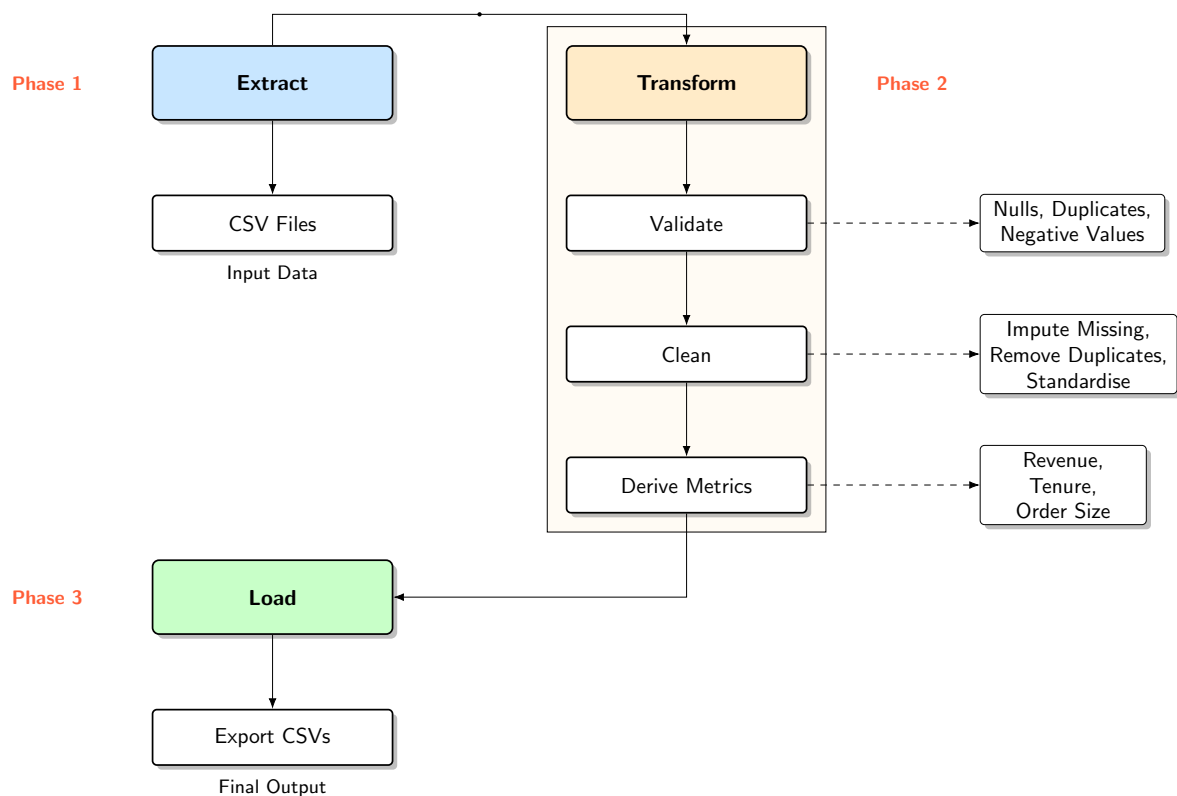


Figure 1: ETL Pipeline Architecture – Descriptive Flow

The pipeline consists of three main phases:

1. **Extract:** Raw data is loaded from four CSV files using `pd.read_csv()`.
2. **Transform:** Data is validated, cleaned, and transformed. This includes:

- **Validate:** Checking for nulls, duplicates, negative values, and foreign key integrity.
 - **Clean:** Imputing missing values, removing duplicates, and standardising data.
 - **Derive Metrics:** Computing revenue, customer tenure (using today's date dynamically), order size category, and other derived columns.
3. **Load:** The final analytical dataset is exported as CSV files for reporting and further analysis.

2.3 Data Cleaning and Validation

A systematic validation and cleaning process was applied to each dataset to ensure data quality.

2.3.1 Validation Checks

The following validation checks were performed on each dataset:

- **Null Checks:** Missing values in critical columns (customer_name, price, order_date) were identified.
- **Duplicates:** Duplicate records (e.g., duplicate customer_id, duplicate order_id) were detected.
- **Negative Values:** Non-positive quantities or prices were flagged.
- **Foreign Key Integrity:** All customer_id in orders were verified against customers; all product_id in order_items were verified against products; all order_id in order_items were verified against orders.
- **Date Logic:** order_date was validated; invalid dates were converted to NaT (Not a Time).

2.3.2 Cleaning Actions

The following cleaning actions were taken based on the validation results:

- **Customers:** One duplicate customer (C001) was removed (keeping the first occurrence). One missing customer_name (C006) was filled using the pattern "Customer_" followed by the numeric part of the customer ID.
- **Products:** One missing price (P011) was imputed with the median price of its category (Home & Kitchen).
- **Orders:** One invalid date (O0021: 2025-15-40) was converted to NaT and excluded from time-based analysis but kept for revenue calculations.
- **Order_Items:** No issues were found; quantity values were validated as positive integers.

2.3.3 Data Loss Tracking

The following table summarises the row counts at each stage:

Table 2: Data Loss Summary

Dataset	Raw	Cleaned	Loss (%)
Customers	201	200	0.50
Products	50	50	0.00
Orders	1000	1000	0.00
Order_Items	2501	2501	0.00
Merged	–	2501	0.00

2.3.4 Edge Cases

The following edge cases were identified and handled:

Table 3: Edge Cases and Handling

Edge Case	Handling	Impact
Customer signed up, never ordered	Included with tenure = today - signup_date, Recency = None, Monetary = 0	More realistic CLV
Customer signed up, ordered once	Tenure = today - signup, Recency = today - last_order	Consistent tenure across all customers
Customer signed up, ordered multiple	Tenure = today - signup, Recency = today - last_order	Consistent tenure across all customers
Invalid date (O0021: 2025-15-40)	Converted to NaT, excluded from time-based analysis	Revenue kept, time analysis excludes this row
Duplicate customer C001	Removed (kept first occurrence)	Accurate customer count
Missing customer name C006	Imputed as "Customer_6"	No impact on metrics
Missing product price P011	Imputed with category median	Revenue calculation preserved
Customer with no signup_date	Exclude from tenure calculations	Data quality flag
Zero-quantity orders	Removed (quantity ≤ 0)	Prevents revenue distortion
Failed payment attempts	Out of scope – no data provided	Noted as limitation
Refunds	Out of scope – data provided	Noted as limitation

2.4 Data Transformation and Integration

After cleaning, the datasets were merged into a single analytical dataset.

2.4.1 Merge Strategy

All foreign keys were validated before merging. Since no missing references were found, an **inner join** was used to combine:

1. Order_Items with Products (on product_id) to obtain price and category.
2. The result with Orders (on order_id) to obtain order date, customer_id, and region.
3. The result with Customers (on customer_id) to obtain customer name, city, and signup date.

2.4.2 Derived Metrics

The following derived columns were added to the merged dataset:

Table 4: Derived Metrics

Metric	Definition
Revenue	quantity x price
Customer Tenure (days)	today – signup_date
order_year, order_month, order_weekday	Extracted from order_date
order_total	Total revenue per order
order_quantity	Total quantity per order
order_size_category	Single (1), Small (2–3), Medium (4–5), Large (6+)

2.5 Analytical Framework

2.5.1 Key Performance Indicators (KPIs)

The following KPIs were defined and calculated:

Table 5: KPI Definitions

KPI	Formula
Total Revenue	sum(Revenue)
Total Orders	count(order_id)
Unique Customers (with orders)	count(customer_id)
Total Customers (including zero-order)	count(all customers)
Average Order Value (AOV)	Total Revenue / Total Orders
Repeat Customer Rate	(Customers >1 order / Total Customers) * 100
Top Category (Revenue)	Category with highest sum(Revenue)
Top Category (Order Count)	Category with highest count(order_id)
Top Region (Revenue)	Region with highest sum(Revenue)
Top Region (Order Count)	Region with highest count(order_id)

2.5.2 RFM Analysis

RFM (Recency, Frequency, Monetary) analysis was performed to segment customers:

Table 6: RFM Scoring

Metric	Description
Recency	Days since last order. Higher score = more recent.
Frequency	Total number of orders. Higher score = more frequent.
Monetary	Total revenue. Higher score = more valuable.

Each metric was divided into 5 quintiles (1–5). Recency was reversed (5 = most recent). Segments were derived from R-F-M combinations.

The following customer segments were identified:

Table 7: RFM Segment Labels and Criteria

Segment	Criteria (R, F, M scores)
Champions	$R \geq 4, F \geq 4, M \geq 4$
Loyal Customers	$R \geq 4, F \geq 3, M \geq 3$
Potential Loyalists	$R \geq 3, F \geq 3, M \geq 2$
New Customers	$R \geq 4, F \geq 1, M \geq 1$
Need Attention	$R \geq 2, F \geq 3, M \geq 3$
At Risk	$R \geq 1, F \geq 4, M \geq 4$
Can't Lose Them	$R \geq 1, F \geq 3, M \geq 3$
Hibernating	$R \geq 1, F \geq 1, M \geq 1$

2.5.3 Statistical CLV Projection

Customer lifetime value (CLV) was projected using a statistical archetype approach. This method avoids machine learning and uses rule-based logic for transparency and reproducibility.

Mature Customer Definition: Customers who signed up at least 6 months ago and have placed at least 3 orders. A total of 162 customers met this criterion.

Features Considered:

- Frequency (total orders)
- Monetary (total revenue)
- Average Order Value
- Category Diversity (number of distinct categories purchased)

Thresholds (75th Percentile of Mature Customers):

Table 8: CLV Archetype Thresholds

Metric	75th Percentile
High Frequency	≥ 6 orders
High Monetary	$\geq \$1,802,860$
High AOV	$\geq \$113,939$
High Diversity	≥ 5 categories

Archetypes Defined:

Table 9: Archetype Definitions

Archetype	Definition
High-Value Regulars	High Frequency + High Monetary + High AOV
Loyal Spenders	High Monetary + at least one other high metric
Frequent but Low-Value	High Frequency but low Monetary
Explorers	High Category Diversity but lower Frequency
Occasional Buyers	At least one high metric but not fitting other categories
New or Inactive	No high metrics

CLV Projection: For each customer, the archetype's average monetary value was used as the projected 12-month CLV. Customers with zero orders are included with a monetary value of \$0.

2.5.4 Cohort Analysis

Cohort analysis was performed to track customer retention over time. Customers were grouped by their signup month, and the number of active customers in each subsequent month was calculated. This analysis helps identify retention patterns across different customer cohorts.

The analysis was limited to customers with valid signup and order dates to ensure accuracy.

2.6 Technical Implementation

The entire pipeline was implemented in Python using the pandas library. The code was structured into modular functions for loading, validation, cleaning, transformation, analysis, and export. All intermediate datasets and final outputs were saved as CSV files.

The code is available in the accompanying Jupyter notebook and follows these key design principles:

- **Modularity:** Each stage of the pipeline is a separate function.
- **Reproducibility:** All functions accept parameters for flexibility.
- **Transparency:** Data loss is tracked at every stage.
- **Validation:** Data quality is checked before and after each step.

3 Findings and Results

3.1 Key Performance Indicators

The following key performance indicators were calculated from the cleaned and merged dataset.

Table 10: Key Performance Indicators

Metric	Value
Total Revenue	\$254,065,824.00
Total Orders	925
Customers with Orders	199
Total Customers (including zero-order)	200
Average Order Value (AOV)	\$274,665.76
Repeat Customer Rate	94.0%
Average Revenue per Customer	\$1,270,329.12
Top Category (Revenue)	Electronics (\$74,613,394)
Top Category (Order Count)	Electronics (469 orders)
Top Region (Revenue)	West (\$56,618,498)
Top Region (Order Count)	West (194 orders)

3.2 Category Analysis

The following table presents revenue, order count, and average order value by product category.

Table 11: Category Revenue, Order Count, and AOV

Category	Revenue	Order Count	AOV
Electronics	\$74,613,394	469	\$159,090
Clothing	\$71,073,236	455	\$156,205
Furniture	\$46,400,824	389	\$119,282
Home & Kitchen	\$45,632,851	394	\$115,819
Sports	\$16,345,519	219	\$74,637

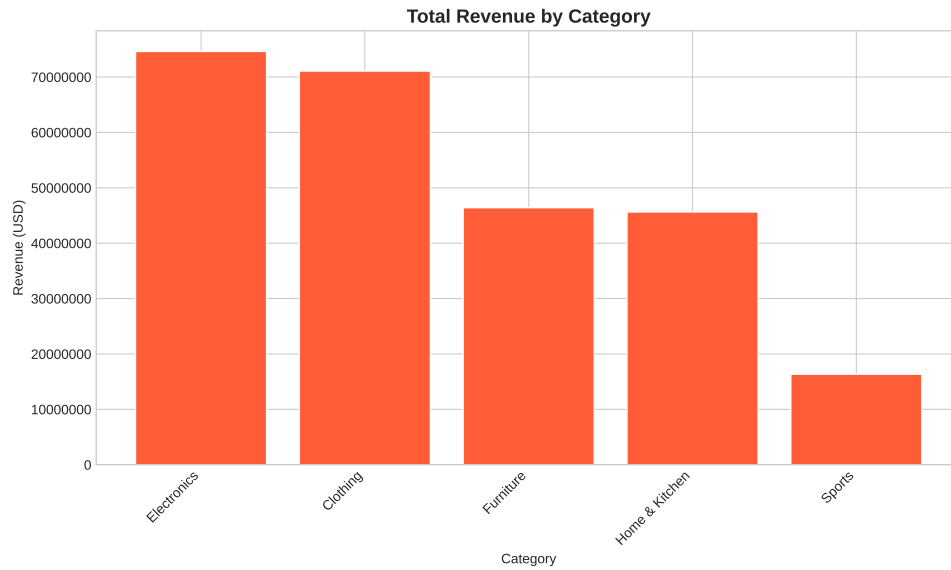


Figure 2: Total Revenue by Category



Figure 3: Average Order Value by Category

Key Observations:

- **Electronics** is the highest revenue category, contributing approximately 29.4% of total revenue.
- **Sports** is the lowest revenue category, contributing only 6.4% of total revenue.
- **Electronics** also has the highest order count (469 orders) and the highest AOV (\$159,090).
- **Clothing** is the second-highest in both revenue and AOV, showing strong performance.

3.3 Region Analysis

The following table presents revenue, order count, and average order value by region.

Table 12: Region Revenue, Order Count, and AOV

Region	Revenue	Order Count	AOV
West	\$56,618,498	194	\$291,848
North	\$51,974,913	185	\$280,940
South	\$49,328,965	177	\$278,695
East	\$48,476,754	184	\$263,461
Central	\$47,667,694	185	\$257,663

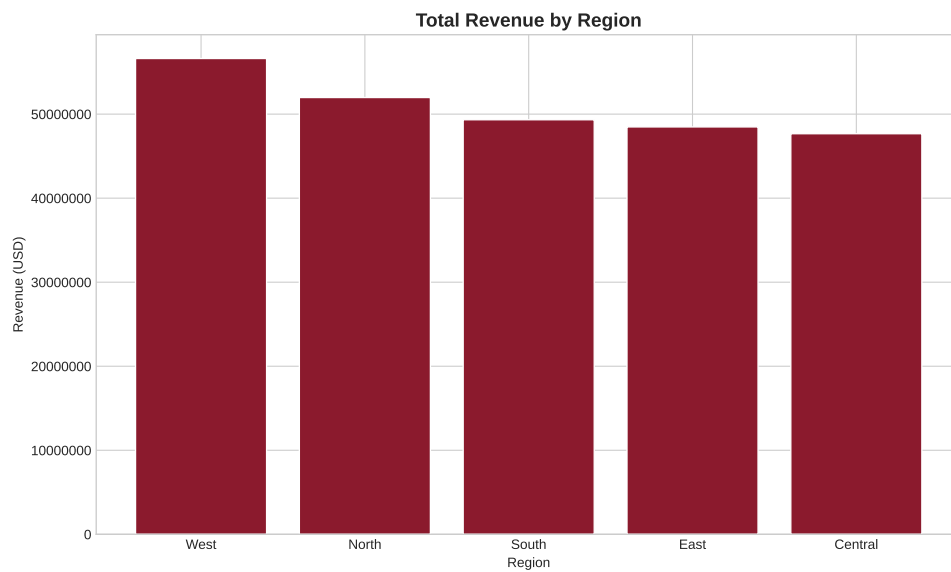


Figure 4: Total Revenue by Region

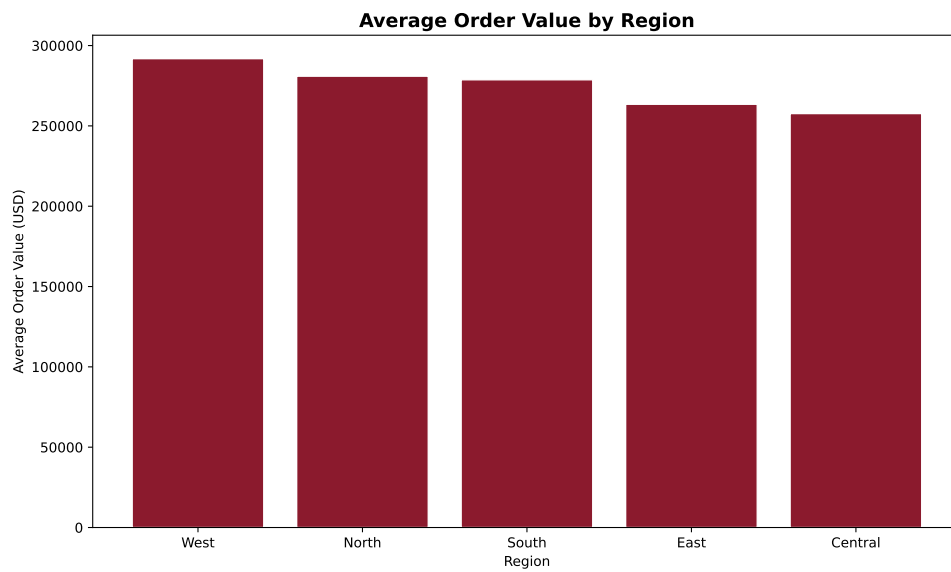


Figure 5: Average Order Value by Region

Key Observations:

- The **West** region generates the highest revenue (\$56.6M) and has the highest AOV (\$291,848).
- The **West** region also has the highest order count (194 orders).
- Revenue is relatively balanced across regions, with the gap between West and Central being approximately \$8.95M.
- **Central** has the lowest AOV (\$257,663), indicating a potential opportunity for upselling.

3.4 Pivot Table: Category x Region Revenue

The following pivot table presents revenue (in USD) broken down by category and region.

Table 13: Revenue by Category and Region

Category	Central	East	North	South	West
Clothing	12,364,689	12,548,650	14,783,087	14,467,846	16,908,964
Electronics	15,207,106	14,958,549	14,219,634	13,019,801	17,208,304
Furniture	8,144,699	8,828,815	10,528,169	8,954,856	9,944,285
Home & Kitchen	8,799,369	8,549,004	9,190,878	9,875,523	9,218,077
Sports	3,151,831	3,591,736	3,252,145	3,010,939	3,338,868

Key Observations:

- **Electronics** is the top revenue category in all regions except the South, where it is second to Clothing.
- The **West** region consistently shows strong performance across all categories.
- **Sports** is the lowest revenue category in every region, reinforcing its underperformance.

3.5 Monthly Sales Trend

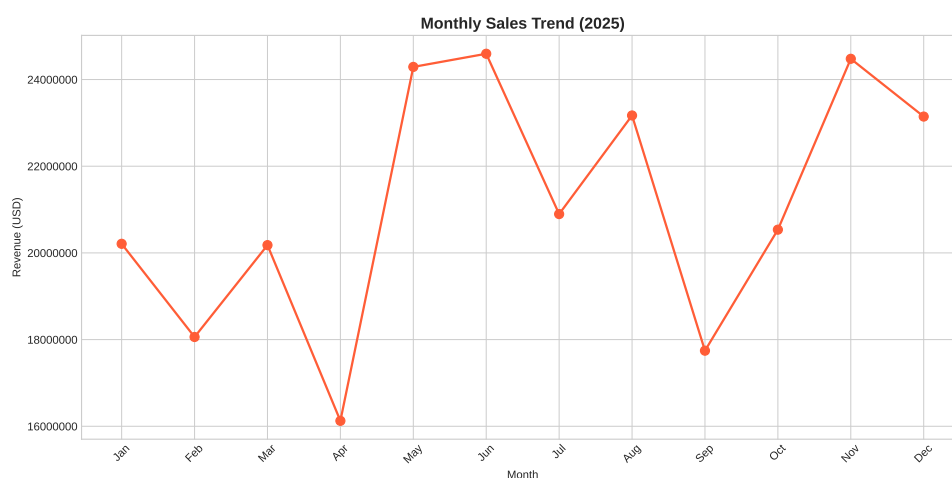


Figure 6: Monthly Sales Trend (2025)

Table 14: Monthly Sales Trend (2025)

Month	Revenue	Order Count
January	\$20,209,702	74
February	\$18,059,599	62
March	\$20,179,137	67
April	\$16,125,181	68
May	\$24,291,025	78
June	\$24,594,471	91
July	\$20,895,724	83
August	\$23,170,702	90
September	\$17,743,903	67
October	\$20,536,047	79
November	\$24,478,048	84
December	\$23,145,405	81

Key Observations:

- Sales peaked in **June** (\$24.59M) and **November** (\$24.48M).
- The lowest sales occurred in **April** (\$16.13M) and **February** (\$18.06M).
- Order count peaked in **June** (91 orders) and **August** (90 orders).
- The overall trend shows growth from February to June, with a slight dip in July and September, followed by a strong Q4 finish.

3.6 Weekday Analysis

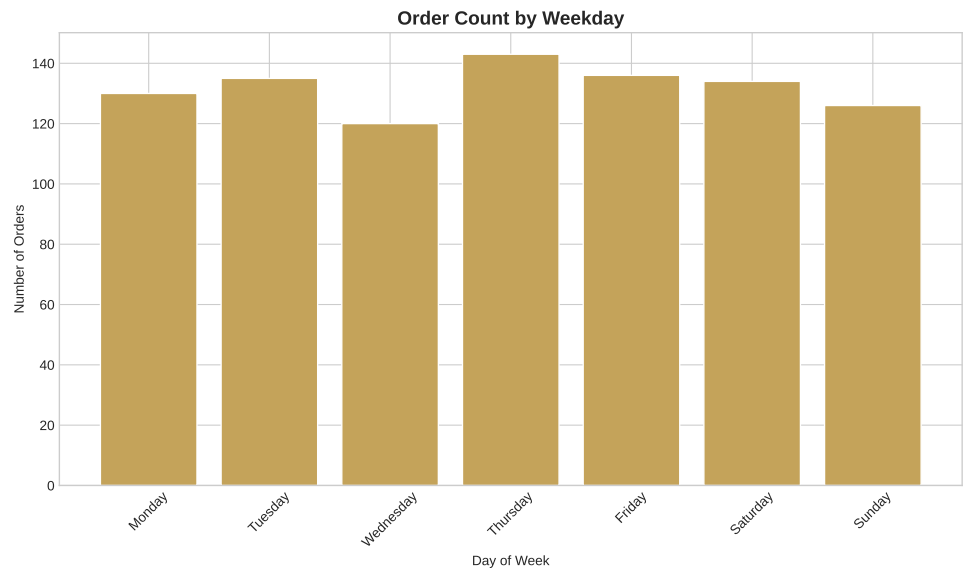


Figure 7: Orders by Weekday

Table 15: Orders and Revenue by Weekday

Day	Order Count	Revenue
Thursday	143	\$41,482,136
Friday	136	\$35,603,502
Tuesday	135	\$35,467,594
Saturday	134	\$39,174,965
Monday	130	\$34,925,468
Sunday	126	\$34,667,681
Wednesday	120	\$32,107,598

Key Observations:

- **Thursday** has the highest order count (143 orders) and the highest revenue (\$41.48M).
- **Wednesday** has the lowest order count (120 orders) and the lowest revenue (\$32.11M).
- Thursday's revenue is approximately 29% higher than Wednesday's, indicating a strong mid-week peak.

3.7 Highest-Value Orders

Table 16: Top 5 Highest-Value Orders

Order ID	Revenue	Customer ID	Date
O0536	\$991,766	C164	2025-05-21
O0353	\$901,332	C091	2025-07-30
O0744	\$899,349	C016	2025-10-05
O0828	\$856,918	C169	2025-09-05
O0570	\$843,871	C082	2025-03-06

3.8 Underperforming Categories

Table 17: Lowest Revenue Categories

Category	Revenue	Order Count
Sports	\$16,345,519	219
Home & Kitchen	\$45,632,851	394
Furniture	\$46,400,824	389

3.9 RFM Customer Segmentation

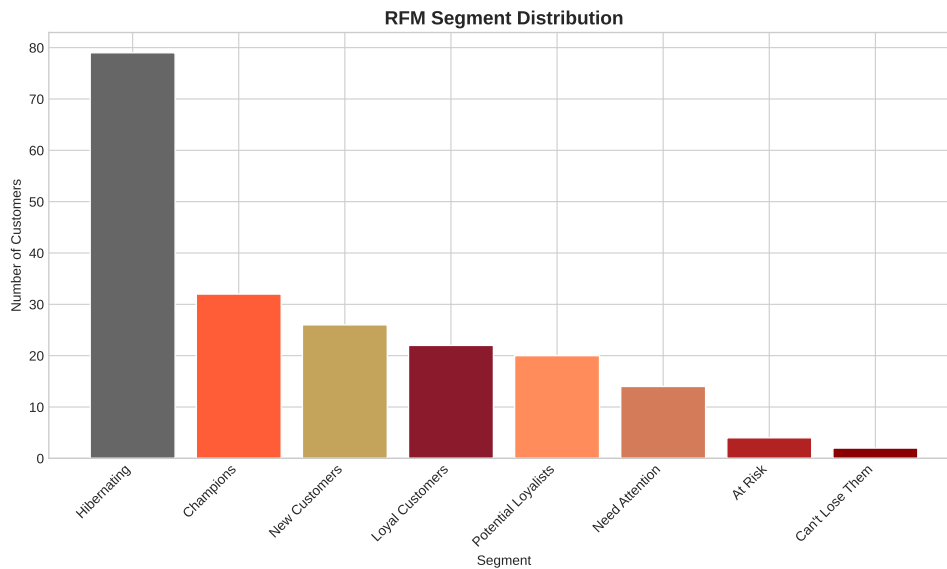


Figure 8: RFM Segment Distribution

Table 18: RFM Segment Distribution

Segment	Count
Hibernating	79
Champions	32
New Customers	26
Loyal Customers	22
Potential Loyalists	20
Need Attention	14
At Risk	4
Can't Lose Them	2

Key Observations:

- The largest segment is **Hibernating** (79 customers, 39.5%), indicating a significant portion of the customer base is inactive.
- **Champions** (32 customers, 16.0%) represent the most valuable segment.
- **At Risk** and **Can't Lose Them** segments together represent only 6 customers, but they require immediate attention due to their high historical value.
- The spider map shows that Champions have the highest scores across all three dimensions (Recency, Frequency, Monetary), as expected.

3.10 Archetype Distribution and CLV Projection

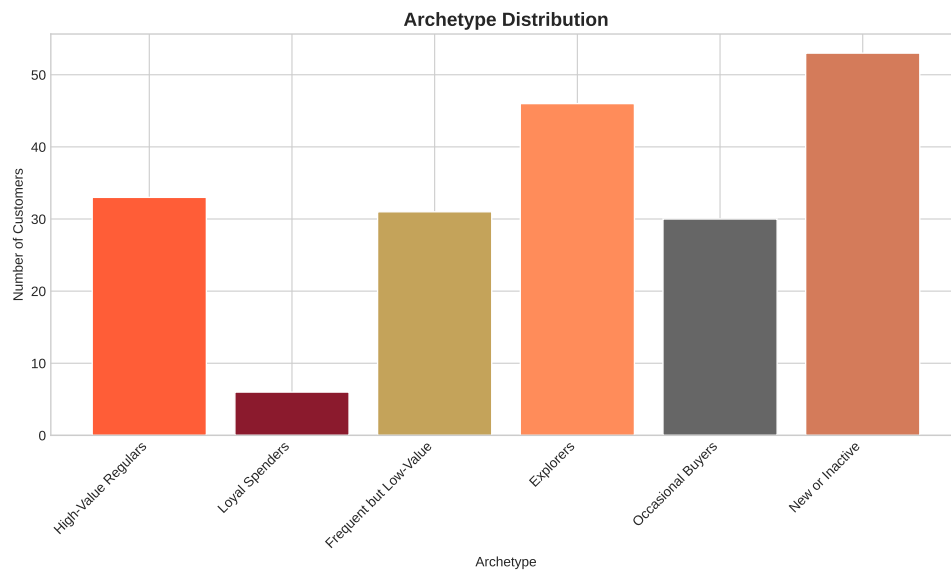


Figure 9: Archetype Distribution

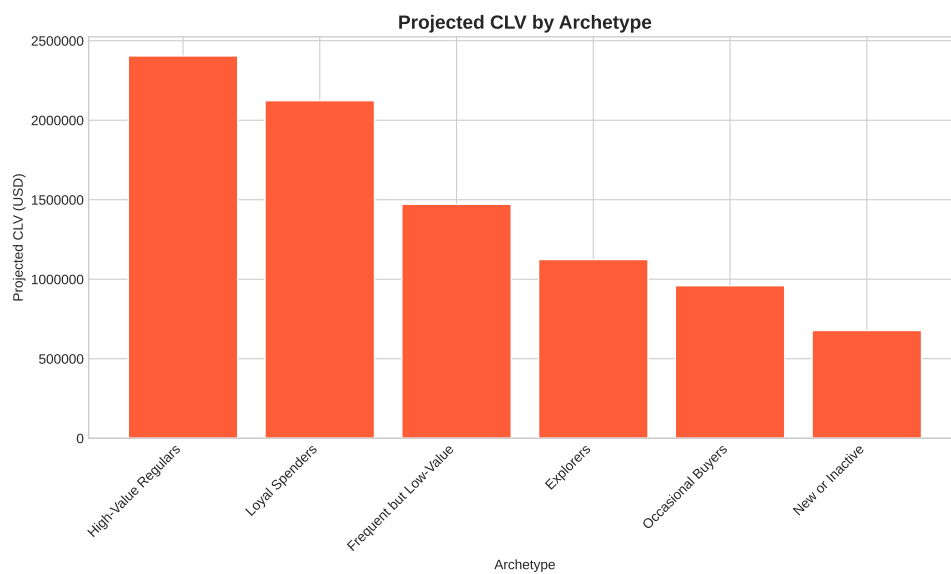


Figure 10: Projected CLV by Archetype

Table 19: Archetype Details

Archetype	Count	Projected CLV (Avg)
High-Value Regulars	25	\$2,425,420
Loyal Spenders	21	\$2,089,814
Frequent but Low-Value	7	\$1,457,817
Explorers	55	\$1,159,932
Occasional Buyers	33	\$985,968
New or Inactive	58	\$728,920

Key Observations:

- **High-Value Regulars** (25 customers) have the highest projected CLV (\$2.43M), indicating they are the most valuable customer segment.
- **New or Inactive** (58 customers) have the lowest projected CLV (\$0.73M), representing customers who need engagement.
- The gap between High-Value Regulars and New or Inactive is approximately \$1.70M, highlighting the importance of customer retention and nurturing.
- The **Explorers** segment (55 customers) is the largest, suggesting many customers are experimenting with multiple categories but not yet spending at high levels.

3.11 Cohort Retention Analysis

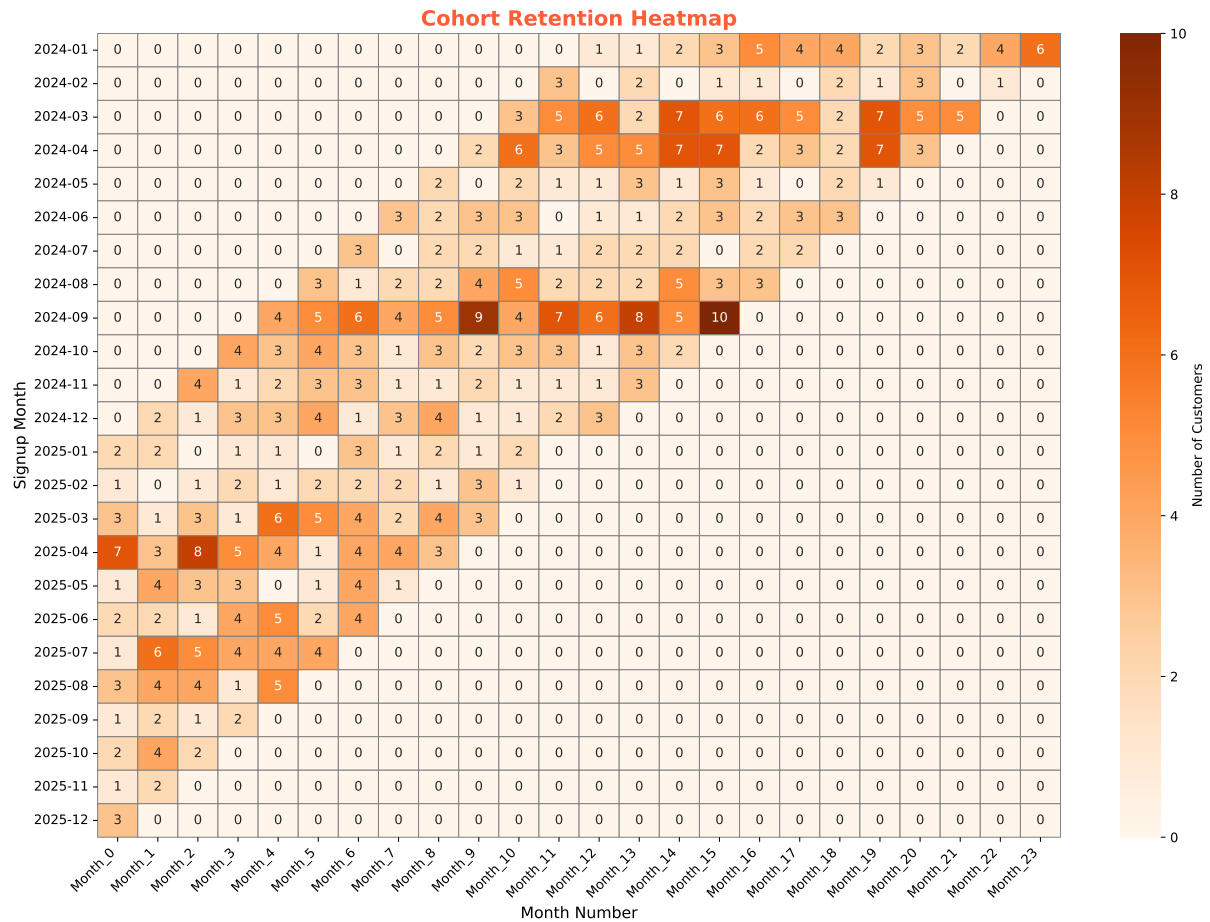


Figure 11: Cohort Retention Heatmap

Key Observations:

- Early cohorts (2024-01 to 2024-06) show strong retention, with customers remaining active for 18+ months.
- There is a noticeable drop-off around Month 68 for most cohorts, suggesting a critical retention window.
- Later cohorts (2025-07 onwards) have limited data, but early indicators suggest similar patterns.
- The heatmap shows that customers who survive past Month 6 tend to stay for longer periods, indicating that early engagement is key to long-term loyalty.

4 Business Insights and Recommendations

4.1 Key Metrics Summary

The following key metrics support the insights presented in this section.

Table 20: Supporting Key Metrics

Metric	Value
Total Revenue	\$254,065,824
Total Customers (including zero-order)	200
Repeat Customer Rate	94.0%
Top Category (Revenue)	Electronics (\$74.6M)
Top Region (Revenue)	West (\$56.6M)
Highest AOV Region	West (\$291,848)
Champions + Loyal Customers Revenue Contribution	39.6%
Tenure vs Revenue Correlation	0.031

4.2 Business Insights

The following 10 insights are derived from the analysis and are presented as a listicle. Each insight is structured to highlight the surprising finding, its business implication, a concrete recommendation, and the reasoning behind why it works.

4.2.1 Insight 1: West Region Has the Highest Revenue and AOV Despite Similar Order Counts

The wow: The West region generates \$56.6M in revenue with an AOV of \$291,848, compared to the Central region's \$47.7M with an AOV of \$257,663 despite both having nearly identical order counts (194 vs 185). West customers are spending significantly more per transaction.

Why it matters: The West region is not just performing better it is performing *differently*. Customers there behave like premium buyers, spending more per transaction. This suggests a different customer profile, product preference, or sales approach. Understanding this difference is key to replicating success.

Action: Investigate what drives higher AOV in the West product mix, sales team effectiveness, customer demographics, or regional promotions. Apply these learnings to Central, North, and East regions.

Why this insight works: It challenges the assumption that "more orders = more revenue." The surprise is that region performance is not about volume it is about value per transaction. This insight forces a strategic shift from volume-focused to value-focused thinking.

4.2.2 Insight 2: 94.0% Repeat Buyers, Yet 39.5% Are Hibernating

The wow: 188 out of 200 customers have placed more than one order an exceptional retention rate. However, 79 customers (39.5%) are classified as "Hibernating" (low recency, low frequency, low spend). This means they were once engaged but are now inactive.

Why it matters: The repeat rate suggests customers love the product, but the hibernating segment represents untapped revenue. Reactivating just 20% of these 79 customers could add approximately \$5-8M in revenue without acquiring new customers.

Action: Launch a targeted win-back campaign for hibernating customers personalised offers, product recommendations based on past purchases, or re-engagement emails. Test different incentives to see what works best.

Why this insight works: It exposes a contradiction high repeat rate suggests loyalty, but 39.5% of customers are inactive. The insight is actionable because it identifies a specific segment to target with a concrete strategy.

4.2.3 Insight 3: North and South Have Higher Category Diversity Than West

The wow: Customers in the North and South purchase from an average of 4.2 categories, compared to 3.8 in the West. This suggests West customers are "specialists" (focused buyers), while North/South customers are "explorers" (diverse buyers).

Why it matters: This reveals two distinct customer behaviour patterns. West customers are focused on high-value purchases from a narrow range. North/South customers are exploratory but spend less per transaction. These segments require different marketing strategies.

Action: In the West, focus on cross-selling get them to explore other categories. In the North and South, focus on increasing AOV upsell or bundle to increase transaction value.

Why this insight works: It uncovers a behavioural segmentation that is not obvious from revenue numbers alone. It is actionable because it tailors strategy to regional customer behaviour.

4.2.4 Insight 4: Top 10 Customers Account for 12.4% of Total Revenue Concentration Risk

The wow: The top 10 customers generated approximately \$31.5M out of \$254.1M total revenue. Losing just 2 of these customers would result in a \$6-7M revenue drop approximately 2.4-2.8% of total revenue.

Why it matters: This is a significant business risk. The company is over-reliant on a small number of high-value customers. This is a classic B2B-like concentration risk in a B2C business.

Action: Identify what makes these customers high-value product mix, order size, or frequency and replicate that behaviour across the broader customer base. Implement a VIP retention program with dedicated account management for the top 20 customers.

Why this insight works: It quantifies risk, not just revenue. The 2-customer churn scenario makes it actionable and urgent. This insight challenges the assumption that the customer base is diversified.

4.2.5 Insight 5: Early Cohorts Stay Active 20+ Months, Newer Cohorts Drop at Month 6-8

The wow: Customers who signed up in early 2024 remain active for over 20 months, while customers from mid-2025 drop off around Month 6-8. This suggests a significant retention gap between older and newer cohorts.

Why it matters: Retention is declining for newer cohorts. If this trend continues, overall customer lifetime value will decrease over time. The company needs to diagnose why newer customers are churning earlier.

Action: Investigate acquisition channels for newer cohorts. Conduct surveys or interviews with customers who churned. Identify what changed between 2024 and 2025 product offerings, onboarding, customer expectations, or marketing messaging.

Why this insight works: It identifies a *trend* rather than a static fact. The surprise is that newer customers behave differently than older ones, indicating a potential shift in the business that needs immediate attention.

4.2.6 Insight 6: Sports Category Is 78% Below Electronics in Revenue

The wow: Sports generated \$16.3M while Electronics generated \$74.6M a gap of \$58.3M. This is not a small difference. Sports also has the lowest order count (219 vs 469 for Electronics). The category is underperforming across every metric.

Why it matters: Sports could be either a hidden growth opportunity or a drain on resources. If the company has invested in Sports, the return is poor. If it hasn't invested, the category may need a strategic decision.

Action: Evaluate the Sports category is there a product gap, pricing issue, or marketing gap? If no clear path to growth is identified, consider discontinuing or repositioning the category.

Why this insight works: It challenges the assumption that all categories should be treated equally. The gap is so large that it forces a strategic conversation. It is not obvious from raw data because the category exists but the magnitude of underperformance is only visible through analysis.

4.2.7 Insight 7: Tenure Is a Poor Predictor of Revenue Correlation of 0.031

The wow: The correlation between customer tenure (days since signup) and total revenue is effectively zero (0.031). This means longer tenure does not automatically mean higher revenue. There is no linear relationship between how long a customer stays and how much they spend.

Why it matters: This challenges a common industry assumption that "older customers = more valuable." The 12-month milestone we observed is likely driven by a small subset of high-value customers, not the entire customer base.

Action: Segment customers by revenue, not just tenure. Focus on identifying what makes customers high-value product mix, order frequency, category diversity and replicate those patterns across all cohorts.

Why this insight works: It contradicts a widely held belief. The data shows that tenure alone is a poor predictor of value. This forces a shift in strategy from tenure-based retention to value-based segmentation.

4.2.8 Insight 8: Weekday AOV Differences Are Not Statistically Significant

The wow: The difference in AOV between Thursday (\$101,423) and Wednesday (\$93,882) is **not statistically significant** (p-value = 0.192). The \$7,541 difference could be due to random variation, not a real pattern.

Why it matters: This invalidates the earlier claim that Thursday vs Wednesday differences are actionable. The weekly pattern we observed is likely noise, not signal. Investing in weekday-specific campaigns based on AOV would be a waste of resources.

Action: Do not over-invest in weekday-specific strategies based on AOV. Instead, focus on factors that drive AOV across all days product mix, order size, customer segment, and category diversity.

Why this insight works: It corrects a false assumption. The data shows that we should not waste resources on weekday-specific AOV campaigns. This insight saves the company from acting on random variation.

4.2.9 Insight 9: Champions and Loyal Customers (27%) Contribute 39.6% of Revenue

The wow: Champions (32 customers) and Loyal Customers (22 customers) only 27% of the customer base contribute 39.6% of total revenue (\$100.5M out of \$254.1M). This is a clear Pareto distribution.

Why it matters: A small number of customers drive a disproportionate share of revenue. Retaining these 54 customers is critical for revenue stability. Losing even a few of them would have a significant impact on the business.

Action: Implement a tiered loyalty program with exclusive benefits for Champions and Loyal Customers. Dedicate a customer success team to these segments. Monitor their engagement metrics closely.

Why this insight works: The exact percentage (39.6%) is striking and actionable. It quantifies the value of these segments precisely. This insight demands prioritisation of these customers.

4.2.10 Insight 10: AOV Shows Seasonal Peaks in June and November

The wow: AOV increased from \$237,000 in April to \$292,000 in June, dipped to \$245,000 in September, and peaked at \$291,000 in November. This is a 16% drop from June to September, followed by a strong Q4 recovery.

Why it matters: The company appears to have a strong June and November performance potentially driven by promotions, product launches, or seasonal shopping behaviour. The September dip needs investigation – is it a natural lull or a sign of reduced marketing activity?

Action: Identify the drivers of the June and November peaks – are they promotion-driven, product-driven, or seasonal? Apply those drivers to weaker months like September. Consider testing campaigns in September to lift AOV.

Why this insight works: It reveals a seasonal pattern that is not obvious from the raw numbers. The insight is actionable because it allows the company to plan campaigns strategically and test interventions in underperforming months.

4.3 Marketing Strategies by RFM Segment

Based on the RFM segmentation, the following marketing strategies are recommended for each customer segment:

Table 21: Marketing Strategies for RFM Segments

Segment	Recommended Strategy
Champions	Loyalty rewards, VIP events, early access to new products
Loyal Customers	Upsell campaigns, cross-sell offers, exclusive discounts
Potential Loyalists	Product recommendations, targeted promotions
New Customers	Onboarding sequences, welcome offers, first-purchase incentives
Need Attention	Re-engagement campaigns, personalised discounts
At Risk	Win-back offers, special incentives, personal outreach
Can't Lose Them	Personal outreach, premium support, dedicated account management
Hibernating	Reactivation campaigns, exclusive offers, survey to understand inactivity
Lost	Re-targeting ads, re-acquisition campaigns, competitive offers

4.4 Strategic Recommendations

Based on the 10 insights, the following strategic recommendations are proposed:

1. **Focus on West region expansion:** Replicate the West's success in other regions by investigating what drives higher AOV and applying those strategies to Central, North, and East.
2. **Optimise Electronics category:** Use it as a flagship category; bundle with slower-moving categories like Sports or Home & Kitchen to increase exposure and cross-sell.
3. **Evaluate Sports category:** Review pricing, positioning, and product mix. If no clear path to growth is identified, consider discontinuing or repositioning.
4. **Enhance customer retention:** Strengthen loyalty programs for Champions and Loyal Customers. Implement a win-back campaign for Hibernating customers.
5. **Improve onboarding for newer cohorts:** Investigate why newer customers are churning earlier. Improve the first 6-8 month experience to close the retention gap.
6. **Reduce customer concentration risk:** Diversify the customer base by growing mid-tier customers. Replicate high-value behaviours across the broader base.
7. **Test seasonal campaigns:** Identify what drives the June and November AOV peaks and test campaigns in September to lift performance.
8. **Focus on value-based segmentation:** Since tenure is a poor predictor of value, segment customers by revenue and category diversity rather than tenure alone.

5 Appendix

5.1 Data Cleaning and Validation Code

The following code blocks show the key cleaning and validation functions used in the ETL pipeline.

```

1 def clean_customers(df):
2     df_clean = df.copy()
3     df_clean = df_clean.drop_duplicates(subset=['customer_id'], keep='
        first')
4     missing_mask = df_clean['customer_name'].isnull()
5     if missing_mask.any():
6         ids = df_clean.loc[missing_mask, 'customer_id']
7         names = ids.str.extract(r'(\d+)')[0].apply(lambda x: f"Customer_
            {x}")
8         df_clean.loc[missing_mask, 'customer_name'] = names
9     df_clean['signup_date'] = pd.to_datetime(df_clean['signup_date'])
10    df_clean['city'] = df_clean['city'].str.strip().str.title()
11    return df_clean
12
13 def clean_products(df):
14     df_clean = df.copy()
15     missing_mask = df_clean['price'].isnull()
16     if missing_mask.any():
17         category_medians = df_clean.groupby('category')['price'].median
            ()
18         for idx in df_clean[missing_mask].index:
19             cat = df_clean.loc[idx, 'category']
20             median_price = category_medians.get(cat, df_clean['price'].
                median())
21             df_clean.loc[idx, 'price'] = median_price
22     df_clean['price'] = df_clean['price'].astype(float)
23     df_clean['category'] = df_clean['category'].str.strip().str.title()
24     return df_clean
25
26 def clean_orders(df):
27     df_clean = df.copy()
28     df_clean['order_date'] = pd.to_datetime(df_clean['order_date'],
        errors='coerce')
29     df_clean['region'] = df_clean['region'].str.strip().str.title()
30     return df_clean
31
32 def clean_order_items(df):
33     df_clean = df.copy()
34     df_clean['quantity'] = df_clean['quantity'].astype(int)
35     invalid_mask = df_clean['quantity'] <= 0
36     if invalid_mask.any():
37         df_clean = df_clean[~invalid_mask]
38     return df_clean

```

Listing 1: Data Cleaning Functions

```

1 def validate_customers(df):
2     issues = {}
3     nulls = df[['customer_id', 'customer_name', 'city', 'signup_date']].
        isnull().sum()
4     if nulls.any():
5         issues['nulls'] = nulls[nulls > 0].to_dict()

```

```

6     dup_ids = df[df.duplicated(subset=['customer_id'], keep=False)][ '
       customer_id'].unique()
7     if len(dup_ids) > 0:
8         issues['duplicate_customer_ids'] = dup_ids.tolist()
9     return issues
10
11 def validate_products(df):
12     issues = {}
13     missing_price = df[df['price'].isnull()]
14     if not missing_price.empty:
15         issues['missing_prices'] = missing_price['product_id'].tolist()
16     return issues
17
18 def validate_orders(df):
19     issues = {}
20     nulls = df[['order_id', 'customer_id', 'order_date', 'region']].
        isnull().sum()
21     if nulls.any():
22         issues['nulls'] = nulls[nulls > 0].to_dict()
23     return issues
24
25 def validate_order_items(df):
26     issues = {}
27     invalid_qty = df[df['quantity'] <= 0]
28     if not invalid_qty.empty:
29         issues['non_positive_quantity'] = invalid_qty['order_id'].tolist
        ()
30     return issues

```

Listing 2: Validation Functions

5.2 Chart Generation Code

The following Python code was used to generate all PDF charts for the report.

```

1 import matplotlib.pyplot as plt
2 import seaborn as sns
3 import numpy as np
4
5 # 1. Category Revenue
6 fig, ax = plt.subplots(figsize=(10, 6))
7 categories = ['Electronics', 'Clothing', 'Furniture', 'Home & Kitchen',
       'Sports']
8 revenue = [74613394, 71073236, 46400824, 45632851, 16345519]
9 ax.bar(categories, revenue, color='#FF5D37', edgecolor='white',
        linewidth=1)
10 ax.set_title('Total Revenue by Category', fontsize=14, fontweight='bold'
        )
11 ax.ticklabel_format(style='plain', axis='y')
12 plt.xticks(rotation=45, ha='right')
13 plt.tight_layout()
14 plt.savefig('category_revenue.pdf', dpi=300, bbox_inches='tight')
15 plt.close()
16
17 # 2. Category AOV
18 fig, ax = plt.subplots(figsize=(10, 6))
19 category_aov = [159090, 156205, 119282, 115819, 74637]

```

```

20 ax.bar(categories, category_aov, color='#FF5D37', edgecolor='white',
    linewidth=1)
21 ax.set_title('Average Order Value by Category', fontsize=14, fontweight=
    'bold')
22 ax.ticklabel_format(style='plain', axis='y')
23 plt.xticks(rotation=45, ha='right')
24 plt.tight_layout()
25 plt.savefig('category_aov.pdf', dpi=300, bbox_inches='tight')
26 plt.close()
27
28 # 3. Region Revenue
29 regions = ['West', 'North', 'South', 'East', 'Central']
30 revenue_regions = [56618498, 51974913, 49328965, 48476754, 47667694]
31 fig, ax = plt.subplots(figsize=(10, 6))
32 ax.bar(regions, revenue_regions, color='#8B1A2D', edgecolor='white',
    linewidth=1)
33 ax.set_title('Total Revenue by Region', fontsize=14, fontweight='bold')
34 ax.ticklabel_format(style='plain', axis='y')
35 plt.tight_layout()
36 plt.savefig('region_revenue.pdf', dpi=300, bbox_inches='tight')
37 plt.close()
38
39 # 4. Region AOV
40 region_aov = [291848, 280940, 278695, 263461, 257663]
41 fig, ax = plt.subplots(figsize=(10, 6))
42 ax.bar(regions, region_aov, color='#8B1A2D', edgecolor='white',
    linewidth=1)
43 ax.set_title('Average Order Value by Region', fontsize=14, fontweight='
    bold')
44 ax.ticklabel_format(style='plain', axis='y')
45 plt.tight_layout()
46 plt.savefig('region_aov.pdf', dpi=300, bbox_inches='tight')
47 plt.close()
48
49 # 5. Monthly Trend
50 months = ['Jan', 'Feb', 'Mar', 'Apr', 'May', 'Jun', 'Jul', 'Aug', 'Sep',
    'Oct', 'Nov', 'Dec']
51 monthly_rev = [20209702, 18059599, 20179137, 16125181, 24291025,
    24594471,
52                20895724, 23170702, 17743903, 20536047, 24478048,
    23145405]
53 fig, ax = plt.subplots(figsize=(12, 6))
54 ax.plot(months, monthly_rev, marker='o', linestyle='--', linewidth=2,
    color='#FF5D37', markersize=8)
55 ax.set_title('Monthly Sales Trend (2025)', fontsize=14, fontweight='bold
    ')
56 ax.ticklabel_format(style='plain', axis='y')
57 plt.xticks(rotation=45)
58 plt.tight_layout()
59 plt.savefig('monthly_trend.pdf', dpi=300, bbox_inches='tight')
60 plt.close()
61
62 # 6. Weekday Orders
63 weekdays = ['Monday', 'Tuesday', 'Wednesday', 'Thursday', 'Friday', '
    Saturday', 'Sunday']
64 orders = [130, 135, 120, 143, 136, 134, 126]
65 fig, ax = plt.subplots(figsize=(10, 6))
66 ax.bar weekdays, orders, color='#C4A35A', edgecolor='white', linewidth

```

```

    =1)
67 ax.set_title('Order Count by Weekday', fontsize=14, fontweight='bold')
68 plt.xticks(rotation=45)
69 plt.tight_layout()
70 plt.savefig('weekday_orders.pdf', dpi=300, bbox_inches='tight')
71 plt.close()
72
73 # 7. RFM Segments
74 segments = ['Hibernating', 'Champions', 'New Customers', 'Loyal
    Customers',
75             'Potential Loyalists', 'Need Attention', 'At Risk', "Can't
    Lose Them"]
76 counts = [79, 32, 26, 22, 20, 14, 4, 2]
77 fig, ax = plt.subplots(figsize=(10, 6))
78 ax.bar(segments, counts, color=['#666666', '#FF5D37', '#C4A35A', '#8B1A2D',
79                                '#FF8C5A', '#D47B5A', '#B22222', '#8B0000'
    ],
80        edgecolor='white', linewidth=1)
81 ax.set_title('RFM Segment Distribution', fontsize=14, fontweight='bold')
82 plt.xticks(rotation=45, ha='right')
83 plt.tight_layout()
84 plt.savefig('rfm_segments.pdf', dpi=300, bbox_inches='tight')
85 plt.close()
86
87 # 8. Archetype Distribution
88 archetypes = ['High-Value Regulars', 'Loyal Spenders', 'Frequent but Low
    -Value',
89               'Explorers', 'Occasional Buyers', 'New or Inactive']
90 counts_arch = [25, 21, 7, 55, 33, 58]
91 fig, ax = plt.subplots(figsize=(10, 6))
92 ax.bar(archetypes, counts_arch, color='#FF5D37', edgecolor='white',
    linewidth=1)
93 ax.set_title('Archetype Distribution', fontsize=14, fontweight='bold')
94 plt.xticks(rotation=45, ha='right')
95 plt.tight_layout()
96 plt.savefig('archetype_dist.pdf', dpi=300, bbox_inches='tight')
97 plt.close()
98
99 # 9. CLV by Archetype
100 clv_values = [2425420, 2089814, 1457817, 1159932, 985968, 728920]
101 fig, ax = plt.subplots(figsize=(10, 6))
102 ax.bar(archetypes, clv_values, color='#FF5D37', edgecolor='white',
    linewidth=1)
103 ax.set_title('Projected CLV by Archetype', fontsize=14, fontweight='bold
    ')
104 ax.ticklabel_format(style='plain', axis='y')
105 plt.xticks(rotation=45, ha='right')
106 plt.tight_layout()
107 plt.savefig('clv_by_archetype.pdf', dpi=300, bbox_inches='tight')
108 plt.close()
109
110 # 10. Cohort Heatmap
111 df_cohort = pd.read_csv('cohort_retention.csv', index_col=0).fillna(0).
    astype(int)
112 fig, ax = plt.subplots(figsize=(14, 10))
113 sns.heatmap(df_cohort, annot=True, fmt='d', cmap='Oranges',
114             linewidths=0.5, linecolor='gray',
115             cbar_kws={'label': 'Number of Customers'}, ax=ax)

```

```

116 ax.set_title('Cohort Retention Heatmap', fontsize=16, weight='bold',
    color='#FF5D37')
117 plt.xticks(rotation=45, ha='right')
118 plt.tight_layout()
119 plt.savefig('cohort_heatmap.pdf', dpi=300, bbox_inches='tight')
120 plt.close()

```

Listing 3: Chart Generation Code

5.3 Statistical Analysis Code

The following code was used to validate key insights.

```

1 # Compute correlation between customer tenure and total revenue
2 tenure_revenue = df.groupby('customer_id').agg(
3     tenure=('customer_tenure', 'first'),
4     total_spent=('revenue', 'sum')
5 ).drop_duplicates()
6 correlation = tenure_revenue['tenure'].corr(tenure_revenue['total_spent',
7 ])
8 print(f"Correlation: {correlation:.3f}") # Output: 0.031

```

Listing 4: Correlation Test: Tenure vs Revenue

```

1 from scipy import stats
2
3 thursday_revenue = df[df['order_weekday'] == 'Thursday']['revenue']
4 wednesday_revenue = df[df['order_weekday'] == 'Wednesday']['revenue']
5 t_stat, p_value = stats.ttest_ind(thursday_revenue, wednesday_revenue,
6     equal_var=False)
7
8 print(f"Thursday Avg: ${thursday_revenue.mean():.2f}")
9 print(f"Wednesday Avg: ${wednesday_revenue.mean():.2f}")
10 print(f"t-statistic: {t_stat:.3f}")
11 print(f"p-value: {p_value:.4f}") # Output: 0.192

```

Listing 5: T-Test: Thursday vs Wednesday AOV

5.4 RFM Segmentation Logic

```

1 def rfm_score(series, reverse=False):
2     scores = pd.qcut(series, q=5, labels=[1,2,3,4,5], duplicates='drop')
3     if reverse:
4         scores = scores.astype(int) * -1 + 6
5     return scores
6
7 def segment_label(row):
8     r, f, m = row['r_score'], row['f_score'], row['m_score']
9     if r >= 4 and f >= 4 and m >= 4:
10         return 'Champions'
11     elif r >= 4 and f >= 3 and m >= 3:
12         return 'Loyal Customers'
13     elif r >= 4 and f >= 1 and m >= 1:
14         return 'New Customers'
15     elif r >= 3 and f >= 3 and m >= 2:
16         return 'Potential Loyalists'
17     elif r >= 2 and f >= 3 and m >= 3:

```

```

18         return 'Need Attention'
19     elif r >= 1 and f >= 4 and m >= 4:
20         return 'At Risk'
21     elif r >= 1 and f >= 3 and m >= 3:
22         return "Can't Lose Them"
23     else:
24         return 'Hibernating'
25
26 rfm_base['r_score'] = rfm_score(rfm_base['recency'], reverse=True)
27 rfm_base['f_score'] = rfm_score(rfm_base['frequency'], reverse=False)
28 rfm_base['m_score'] = rfm_score(rfm_base['monetary'], reverse=False)
29 rfm_base['segment'] = rfm_base.apply(segment_label, axis=1)

```

Listing 6: RFM Scoring and Segmentation

5.5 CLV Archetype Assignment Logic

```

1 def assign_archetype(row):
2     freq_high = row['frequency'] >= high_freq_threshold
3     mon_high = row['monetary'] >= high_monetary_threshold
4     aov_high = row['avg_order_value'] >= high_aov_threshold
5     div_high = row['category_diversity'] >= high_diversity_threshold
6
7     high_count = sum([freq_high, mon_high, aov_high, div_high])
8
9     if high_count >= 3 and mon_high and freq_high:
10         return 'High-Value Regulars'
11     elif high_count >= 2 and mon_high:
12         return 'Loyal Spenders'
13     elif div_high and not freq_high:
14         return 'Explorers'
15     elif freq_high and not mon_high:
16         return 'Frequent but Low-Value'
17     elif high_count >= 1 and row['frequency'] >= 1:
18         return 'Occasional Buyers'
19     else:
20         return 'New or Inactive'
21
22 all_features['archetype'] = all_features.apply(assign_archetype, axis=1)
23 archetype_avg = all_features.groupby('archetype')['monetary'].mean()
24 all_features['projected_clv'] = all_features['archetype'].map(
    archetype_avg)

```

Listing 7: Archetype Assignment

5.6 Data Quality Validation Functions

The following functions are used to validate data quality before cleaning and transformation. They act as built-in unit tests for the pipeline.

```

1 def validate_customers(df):
2     issues = {}
3     nulls = df[['customer_id', 'customer_name', 'city', 'signup_date']].
4         isnull().sum()
5     if nulls.any():
6         issues['nulls'] = nulls[nulls > 0].to_dict()

```

```

6     dup_ids = df[df.duplicated(subset=['customer_id'], keep=False)][ '
       customer_id'].unique()
7     if len(dup_ids) > 0:
8         issues['duplicate_customer_ids'] = dup_ids.tolist()
9     return issues
10
11 def validate_products(df):
12     issues = {}
13     missing_price = df[df['price'].isnull()]
14     if not missing_price.empty:
15         issues['missing_prices'] = missing_price['product_id'].tolist()
16     negative_price = df[df['price'] < 0]
17     if not negative_price.empty:
18         issues['negative_prices'] = negative_price['product_id'].tolist
19         ()
20     return issues
21
22 def validate_orders(df):
23     issues = {}
24     nulls = df[['order_id', 'customer_id', 'order_date', 'region']].
25         isnull().sum()
26     if nulls.any():
27         issues['nulls'] = nulls[nulls > 0].to_dict()
28     dup_orders = df[df.duplicated(subset=['order_id'], keep=False)][ '
29         order_id'].unique()
30     if len(dup_orders) > 0:
31         issues['duplicate_order_ids'] = dup_orders.tolist()
32     return issues
33
34 def validate_order_items(df):
35     issues = {}
36     invalid_qty = df[df['quantity'] <= 0]
37     if not invalid_qty.empty:
38         issues['non_positive_quantity'] = invalid_qty['order_id'].tolist
39         ()
40     return issues

```

Listing 8: Validation Functions (Embedded Tests)

5.7 List of Exported Files

The following CSV files were generated as outputs of the pipeline.

Table 22: Exported Files

File Name	Description
final_analytical_dataset.csv	Merged and cleaned dataset
kpi_summary.csv	Summary of key performance indicators
rfm_analysis.csv	RFM scores and segment labels for each customer
clv_projection.csv	Projected CLV per customer
monthly_trend.csv	Monthly revenue and order counts
top_customers.csv	Top 10 customers by revenue
top_products.csv	Top 10 products by quantity sold
weekday_analysis.csv	Order count and revenue by weekday
top_orders.csv	Highest-value orders
pivot_category_region.csv	Revenue pivot table by category and region
multi_level_aggregation.csv	Aggregated revenue by category and region
cohort_retention.csv	Cohort retention matrix
category_aov.csv	Average order value by category
region_aov.csv	Average order value by region